

Mathematical Foundations

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1 Analysis

Analysis I

1. Real Numbers and the completeness
2. Limits and continuity in metric space
3. Open and closed sets in metric space
4. Topological space & The closure of the set
5. Equivalence of continuous functions & Inheritance Principle & Homeomorphism
6. Product metric spaces & Completeness & Compactness (sequentially compact)
7. Compactness (2) & Bolzano-Weierstrass Thm & Nested seq of nonempty compact sets
8. Continuous functions and Homeomorphisms on compact sets & Connectedness
9. Some properties in topological spaces
10. Accumulation points and isolated points & Dense & Hausdorff space
11. Properties of Connectedness
12. Path connected & Compactness in metric space, Hausdorff space, Topological space(1)
13. Compactness (2) & Sequentially compact implies compact(open cover) (1)
14. Sequentially compact implies compact(open cover) (2) & Totally bounded (1)
15. Totally bounded (2) & Perfect & Cantor set
16. Cantor surjection Theorem
17. Peano Curves & Riemann integrable & Darboux integrable(1)
18. Darboux integrable(2) & Riemann integrable iff Darboux integrable(1)
19. Riemann integrable iff Darboux integrable(2) & Some integrable functions (1)
20. Some integrable functions (2) & Lebesgue number & Zero set
21. Oscillation & Riemann-Lebesgue Theorem
22. Some applications of R-L theorem
23. Fundamental Theorem of Calculus & Convergence in functional space

24. Complete functional space & Convergence function series & Integrals wst unif conv.
25. Derivatives w.s.t uniform convergence & Equicontinuity and Arzela-Ascoli Theorem
26. Heine-Borel Theorem in a function space & Stone-Weierstrass Theorem
27. Fixed point, contraction and ODE
28. Nowhere differentiable continuous function & Baire' s Theorem
29. Corollary of Baire' s Theorem
30. Multiple integral and Fubini's Theorem
31. Change of variables formula (1)
32. Change of variables formula (2)
33. Banach spaces
34. Banach algebra
35. Differentiation in Banach Spaces
36. Some examples & Integration in Banach spaces (1)
37. Integration in Banach spaces (2)
38. Inverse function theorem
39. Lebesgue Outer measure
40. Lebesgue measurable set & Abstract outer measure
41. Measurable Sets form Sigma-Algebra & Measure continuity
42. Regularity (1)
43. Regularity (2)
44. Every open set is a countable disjoint union of balls plus a zero set
45. Measure Product & Inner measure (1)
46. Inner measure (2) & Slice Measure (1)
47. Slice Measure (2) & Lebesgue integrable functions & Monotone Convergence Theorem
48. Completed undergraph & Dominated Convergence Thm and Fatou' s Lem & Basic properties

- 49. [Some basic properties of measurable functions \(2\)](#)
- 50. [Cavalieri's principle & Equivalence measurable function definition & Tonelli Thm](#)
- 51. [Vitali covering Lemma](#)
- 52. [Density Theorem](#)

Analysis II

1. Lebesgue Differentiation Theorem (1)
2. Lebesgue Differentiation Theorem (2) & Absolutely continuous functions
3. Indefinite integral is abs conti. & A monotone function has a derivative a.e. (1)
4. A monotone function has a derivative almost everywhere (2)
5. Riemann integrals vs Lebesgue integrals
6. Bounded variation
7. Equi relation of B-V functions and abs conti functions & R-S integrals (1)
8. Riemann-Stieltjes integral (2)
9. Riemann-Stieltjes integral (3)
10. Riemann-Stieltjes integral (4)
11. $L^2 [a, b]$ space
12. Properties of $L^2 [a, b]$ space
13. Orthogonal basis in L^2
14. Fourier coefficients and Parseval's formula
15. A necessary condition of Fourier series converges in L^1 & Riemann-Lebesgue lemma
16. Dirichlet kernel
17. Dini Theorem & There is a continuous function whose Fourier series diverges (1)
18. There is a continuous function whose Fourier series diverges (2) & U-B Principle
19. There is a continuous function whose Fourier series diverges (3)
20. Some properties of Fourier coefficients & Cesàro summability
21. Fejér's Theorem & Approximate integrable functions by tri-polynomial in L^1 norm
22. The average of Fourier series converges in L^p norm
23. Equi. of the average of Fourier series converges unif & Weakly convergence(1)
24. Weakly convergent(2) & Equi condition for tri-series to be the Fourier series

25. Convergence of Cesàro mean of Fourier series (1)
26. Part2 Conv of Cesàro mean of Fourier series(2), Conjugate Fourier series & Dini theorem
27. Singular integrals and conjugate function
28. Bound of the L^2 norm of conjugate function & Hardy-Littlewood Maximal function
29. Bound of the norm of the supremum of conjugate functions
30. A counterexample & C-Z Decomposition & The conjugate function exists a.e. in L^1
31. The conjugate function exists a.e. in L^1 (2)
32. The conjugate function exists a.e. in L^1 (3) & Weakly estimate
33. Applications of Fourier series -isoperimetric inequality
34. Applications of Fourier series -continuous but nowhere differentiable function
35. Fourier transform and some properties
36. Fourier inversion formula and Minkowski's inequality
37. Poisson summation formula & Roth's theorem on 3-term AP
38. Discrete Fourier series, Fourier transform and some properties (1)
39. Discrete Fourier series, Fourier transform and some properties (2)
40. Bourgain's theorem
41. Proof of Roth's theorem on 3-terms AP (step 1)
42. Proof of Roth's theorem on 3-terms AP (step 2 to step 7)
43. Proof of Roth's theorem on 3-terms AP (step 9 , step 8)
44. Proof of Roth's theorem on 3-terms AP (step 8)
45. There is f in L^1 such that for a.e. x , the Fourier series diverges (1)
46. There is f in L^1 such that for a.e. x , the Fourier series diverges (2)
47. There is f in L^1 such that for a.e. x , the Fourier series diverges (3)
48. There is f in L^1 such that for a.e. x , the Fourier series diverges (4)

2 Calculus

Calculus I

1. System of real number 1
2. System of real number 2
3. System of real number 3
4. Completeness of Real Number, Weierstrass Theorem 1
5. Weierstrass Theorem 2, define subsequence, Cauchy Sequence
6. Least Upper Bound & Greatest Lower Bound
7. Continuous function, uniformly continuous, Lipchitz continuous
8. Intermediate value theorem, Extreme Value Theory
9. Rieman sum
10. Properties of integral, MVT/general MVT for integral, logarithm
11. Logarithm and exponential 1
12. Logarithm and exponential 2
13. Fundamental Theorem of Calculus 1, Rolle's Theorem, MVT for derivative 1
14. MVT for derivative 2, approximation by linear function, FTC 2
15. Properties of derivative, composite functions, chain rule
16. First/second derivative test
17. Order of magnitude, introduce an smooth function
18. Some example of oscillating functions, chain rule, method of substitution
19. Integration by parts
20. Integration of rational functions 1
21. Integration of rational functions 2, Improper integral 1
22. Improper integral 2
23. Taylor's series: Gamma function, power series, expansion of the logarithm

24. Taylor's series: Taylor's theorem, Cauchy's & Lagrange's remainder
25. Taylor's series: estimate remainder term, expansion of elementary
26. Taylor's series: examples
27. Taylor's polynomial: interpolation 1
28. Taylor's polynomial: interpolation 2, approximation in first order
29. Taylor's polynomial: approximation in second order
30. Taylor's polynomial: fixed point approximation, Stirling's formula
31. Series: concepts of convergence and divergence 1
32. Series: rearrangement, tests for absolute convergence and divergence 1
33. Series: tests for abs. convergence and divergence 2, sequences of functions
34. Series: pointwise/uniform convergence 1
35. Series: pointwise/uniform convergence 2
36. Series: power series, interval of convergence
37. Series: product of two power series
38. Series: expansion of given power series, infinite product
39. Fourier series: periodic function, complex notation 1
40. Fourier series: complex form 2, trigonometric formula, Riemann Lebesgue Lemma
41. Fourier series: examples of Fourier expansion
42. Fourier series: main theorem on Fourier expansion
43. Fourier series: examples of Fourier series 1
44. Fourier series: examples of Fourier series 2, Bessel's Inequality
45. Approximation by trigonometric and rational polynomial 1
46. Approximation by trigonometric and rational polynomial 2
47. Approximation by trigonometric and rational polynomial 3
48. Inner product

- 49. [Bernoulli polynomial and their applications 1](#)
- 50. [Bernoulli polynomial and their applications 2](#)

Calculus II

1. Functions of multiple variables & partial derivative
2. Continuity
3. Differentiability & directional derivative (1)
4. Directional derivative (2) & tangent plane
5. Change of variables & Taylor series
6. MVT & Taylor expansion & Integral
7. Double integral & length of curve
8. Curvature & linear differential one form
9. Line integral
10. Heine-Borel theorem
11. Compact subset & Implicit Function theorem (1)
12. Implicit Function theorem (2)
13. Inverse Function theorem (1)
14. Inverse Function theorem (2) & extreme value
15. Lagrange Multiplier method
16. Examples & Hölder Inequality
17. sufficient condition for local max/min
18. Jordan measurable
19. Jordan area (1)
20. Jordan area (2)
21. Double integral (1)
22. Double integral (2)
23. Transformation of multiple integrals
24. Improper multiple integrals (1)

25. Improper multiple integrals (2) & volumes
26. Surface area
27. Surface area formula
28. Multiple integral in curved coordinate
29. Extend to $\mathbb{R}^n$
30. Integral of unbounded set
31. Fourier integral theorem (1)
32. Fourier integral theorem (2)
33. Fourier transform - decay property & Parseval's identity
34. Fourier transform of several variables
35. Green theorem
36. Divergence theorem
37. Prove Inverse Function theorem by Green theorem
38. Orientation of surface in $\mathbb{R}^3$ I
39. Orientation of surface in $\mathbb{R}^3$ II
40. Gauss theorem in $\mathbb{R}^3$ I
41. Gauss theorem in $\mathbb{R}^3$ II
42. Application of fluid
43. Stoke's theorem
44. General surface (partition of unit) (1)
45. General surface (partition of unit) (2)
46. General surface (partition of unit) (3) & functions of one complex variable
47. Complex function - power series
48. Complex function - differentiability
49. Conformal map & integration of analytic functions - Cauchy's theorem

- 50. [Complex integral theorem & Cauchy integral formula](#)
- 51. [Zeros, poles and residues](#)

Orthogonal Curvilinear Coordinates

3 Set Theory

4 Lie Algebra

1. Introduction to Lie Groups with Focus on Matrix Groups A
2. Introduction to Lie Groups with Focus on Matrix Groups B
3. Introduction to Lie Groups with Focus on Matrix Groups C
4. Definition, Examples, Lie Subalgebra, Ideals, Adjoint Homomorphism
5. Adjoint Homomorphism, Structure Constants, Cross Product Lie Algebra
6. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical A
7. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical B
8. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical C
9. Solvable, Nilpotent and Simple Lie Algebras A
10. Solvable, Nilpotent and Simple Lie Algebras B
11. Lemmas, Engel and Lie's Theorems, Lie Algebra Homomorphism, Modules A
12. Lemmas, Engel and Lie's Theorems, Lie Algebra Homomorphism, Modules B
13. Representations, Factor Modules, Irreducible Submodules A
14. Representations, Factor Modules, Irreducible Submodules B
15. Schur's Lemma, $SL(2, \mathbb{C})$ Representation on V_d Begins
16. $SL(2, \mathbb{C})$ Representation on V_d Continued, Irreducibility Confirmed

5 Linear Algebra

Linear Algebra I

1. Fields, Vector Spaces, Subspaces.
2. Subspaces, Linear Dependence, Basis.
3. Linear Dependence, Basis.
4. Basis, Replacement Theorem, Dimension.
5. Linear Transformation, Kernel, Range, Dimension Theorem.
6. Dimension Theorem, Projection.
7. Matrix Representations.
8. Matrix Representations, Invertible Linear Transformations.
9. Invertible Linear Transformations, Change of Coordinates.
10. Elementary Matrices.
11. Elementary Matrices, Gaussian Elimination.
12. Gaussian Elimination, Determinant of Order 2.
13. Determinants.
14. Determinants(in Terms of Permutations),Cramer's Rule,Adjoint Matrices.
15. Diagonalization
16. Diagonalization
17. Invariant Subspaces, Cayley-Hamilton Theorem
18. Jordan forms, generalized eigenspaces (w. notes)
19. Jordan forms
20. Jordan forms.
21. Jordan forms.
22. Exponential of matrices
23. Systems of first order differential equations, minimal polynomials.

24. Rational canonical forms.
25. Rational canonical forms.
26. Rational canonical forms.

Linear Algebra II

1. Dual Spaces
2. Quotient Spaces
3. Inner Product Space
4. Inner Product Space, Gram-Schmidt Process
5. Orthogonal Projection
6. Hilbert Space, Riesz Representation Theorem
7. Riesz Representation Theorem
8. Adjoint Operators, Normal Operators
9. Normal and Self adjoint Operators
10. Positive semidefinite, Orthogonal and Unitary operators
11. Orthogonal and Unitary operators
12. Orthogonal projection, Spectral theorem
13. Spectral theorem, Simultaneous diagonalization of normal operators
14. Singular value decomposition, Polar decomposition
15. Pseudoinverse
16. Pairs of commuting matrices over a finite field
17. Pairs of commuting matrices over a finite field
18. Bilinear forms
19. Bilinear forms, Quadratic forms
20. Quadratic forms
21. Quadratic spaces
22. Quadratic spaces
23. Witt decomposition theorem
24. Cartan-Dieudonné theorem

- 25. [Cartan-Dieudonné theorem](#)
- 26. [Bézout's theorem](#)
- 27. [Bézout's theorem](#)

6 Basic Geometry

1. Course Overview and a Brief History of Geometry
2. A Brief History of Geometry Continued and Euclid's Axiomatic System
3. Hilbert's Axiomatic System
4. Reflections and Translations
5. Vector Spaces
6. Euclidean Spaces
7. Bases and Orientations
8. Exterior Products
9. An Introduction to Spherical Geometry
10. Models of Hilbert's Axioms
11. Equations of Planes and Lines
12. Curves and Surfaces
13. Geometric Transformations
14. Rigid Transformations
15. Equivalence Classes and Classification of Conic Sections
16. Classification of Quadric Surfaces
17. The Erlangen Program
18. A Comparison of Plane Geometries
19. The Projective Plane
20. Lines in the Projective Plane
21. Projective Transformations
22. The Cross Ratio
23. Characterizing Projective Transformations of the Projective Plane
24. Applications of Projective Transformations

- 25. An Intuitive Introduction to Topology
- 26. Euler's Formula for Polyhedra
- 27. Topological Surfaces

7 Number Theory

1. Introduction
2. Survey
3. Divisibility and Euclid's algorithms
4. More on Euclid's algorithm
5. Primes
6. Multiplicative functions
7. Binomial coefficients
8. Applications of binomial coefficients
9. Congruences
10. Fermat's theorem
11. Euler's theorem
12. Wilson's theorem
13. The Chinese remainder theorem
14. Euler's totient function
15. Numerical calculation
16. More numerical calculation
17. Factorization
18. Cryptography
19. Hensel and Newton's method
20. p -adic numbers
21. Congruences modulo a prime
22. Chevalley-Waring theorem
23. Primitive roots
24. Primitive roots for prime powers
25. Quadratic equations mod p
26. Roots of polynomials modulo a prime
27. Groups and number theory
28. Products of groups
29. Rings in number theory
30. Fields in number theory
31. Quadratic residues
32. Calculation of the Legendre symbol
33. Quadratic reciprocity
34. Gauss sums
35. Jacobi symbol
36. Kronecker symbol
37. Continued fractions
38. Binary quadratic forms
39. Equivalence of binary quadratic forms
40. Examples of positive definite forms
41. More examples of binary quadratic forms
42. Examples of indefinite binary quadratic forms
43. Gaussian integers
44. Pythagorean triangles
45. Dirichlet series
46. Products of Dirichlet series
47. The prime number theorem
48. Proof of the prime number theorem
49. Dirichlet's theorem

50. Dirichlet characters

51. Proof of Dirichlet's theorem

52. Nonvanishing of L series at $s = 1$

53. Three calculators for number theorists

8 Measure Theory

1. [Introduction: a Non-Measurable Set](#)
2. [Classes of Subsets \(Semi-Algebras, Algebras and Sigma-Algebras\), and Set Functions](#)
3. [Set Functions](#)
4. [Caratheodory Theorem](#)
5. [Monotone Classes](#)
6. [The Lebesgue Measure I](#)
7. [The Lebesgue Measure II](#)
8. [Complete Measures](#)
9. [Approximation Theorems](#)
10. [Integration: Measurable and Simple Functions](#)
11. [Measurable Functions](#)
12. [Definition of the Integral](#)
13. [Integral of Simple Functions](#)
14. [Properties of the Integral I](#)
15. [Properties of the Integral II](#)
16. [Theorems on the Convergence of Integrals](#)
17. [Product Measures](#)
18. [Measure on a Countable Product of Spaces](#)
19. [Fubini's Theorem](#)
20. [Hahn-Jordan Theorem](#)
21. [Radon-Nikodym Theorem](#)
22. [Almost Sure and Almost Uniform](#)
23. [Convergence in Measure](#)
24. [Hölder and Minkowski Inequalities](#)

- 25. L_p Spaces
- 26. From Convergence in Measure to Convergence in L_p
- 27. Bounded Linear Operators in L_p
- 28. Vitali's Covering Lemma
- 29. Differentiability of Functions of Bounded Variations
- 30. Absolutely Continuous Functions
- 31. Decomposition of Distribution
- 32. Cantor Ternary Set and Function
- 33. Materials

9 Abstract Algebra

Abstract Algebra I

1. Basic notion, Symmetry groups
2. Cosets, Normal Subgroups, Quotient Groups, Isomorphism Theorem
3. Isomorphism Theorem, Tower of Subgroups
4. Butterfly lemma, Jordan-Hölder theorem
5. Jordan-Hölder theorem, Cyclic groups
6. Group Action, Simplicity of A_n
7. Simplicity of A_n , Sylow theorems
8. Sylow theorems
9. Semidirect products
10. Structure theorem of finite abelian groups
11. Ring Isomorphism Theorem
12. Chinese Remainder Theorem, Localization
13. ED, PID, UFD
14. Gauss lemma, Criteria for irreducibility
15. Algebraic extensions
16. Algebraic closures
17. Algebraic closures, Normal extensions
18. Separable extensions
19. Primitive element theorem, Finite fields
20. Galois theory
21. Galois theory
22. Galois theory, roots of unity, cyclotomic extensions.
23. Norms and traces.

- 24. Purely inseparable extensions, cyclic extensions.
- 25. Solvable extensions, solvable by radicals.

Abstract Algebra II

1. Module theory.
2. Free modules, projective modules.
3. Modules over PID.
4. Modules over PID.
5. Tensor products.
6. Modules of fractions.
7. Noetherian modules.
8. Primary decompositions.
9. Primary decompositions, Nakayama's lemma.
10. Nakayama's lemma, filtered and graded modules, Artin-Rees.
11. Hilbert polynomial, Artinian modules.
12. Abelian categories, general homology theory.
13. Injective modules.
14. Homotopies of morphisms of complexes, derived functors.
15. Ext functor.
16. Ext functor, Ext^1 , Tor functor.
17. Group cohomology.
18. Group cohomology, Semisimple modules.
19. Semisimple modules, Semisimple rings.
20. Density theorem, Wedderburn theorem, Burnside theorem.
21. Representations, Characters.
22. Characters, Class functions.
23. Orthogonality relations.
24. Orthogonality relations.

25. Induced characters, Induced representations.
26. Induced representations, Mackey's criterion.
27. Artin theorem, Brauer theorem.

10 General Topology

1. Metric Spaces and Topologies: Open Sets, Equivalence, and Metrizability
2. Closed Sets and Continuity: Metric Definitions, Subspaces, Bases, Topology
3. Continuous Maps, Subbases, Generated Topologies, and Cartesian Products
4. Product Topologies, Homeomorphisms, Continuous Bijections, and Uniqueness
5. Products, Universal Properties, Box Topology, Open/Closed Maps, and Coproducts
6. Quotient Topology and Limit Points
7. Closure, Limit Points, Sequence Convergence, and Nonunique Limits
8. Interior, Boundary, Closure, Complements, and First-Countable Topological Spaces
9. Preorders, Directed Sets, Nets, Convergence, Continuity, and Hausdorff Spaces
10. Subnets, Compactness, Hausdorff Spaces, and Homeomorphism Criteria
11. Compactness, Bolzano-Weierstrass, Tube Lemma, Heine-Borel, and Extreme Values
12. Finite Intersection Property, Net Cluster Points, Subnets, and Compactness
13. Tychonoff's Theorem, Product Compactness, and the Lebesgue Number Lemma
14. Metric Compactness, Sequential Compactness, Completeness, Total Boundedness, and Separation Axioms
15. Regularity, Normality, Hausdorff Spaces, Closed Neighborhoods, and Metric Spaces
16. Compact Hausdorff Spaces, Complete Regularity, Tychonoff Spaces, and Urysohn's Lemma
17. Metrizability of $[0, 1]^N$ and Urysohn's Metrization Theorem
18. Second Countability, Metrizability, Lindelöf Spaces, and the Tietze Extension Theorem
19. Moore Plane, Local Compactness, Manifolds, LCH Spaces, and Compactifications
20. One-Point Compactifications, LCH Spaces, Uniqueness, Proper Maps, and Extensions
21. Proper Maps, One-Point Compactifications, Topological Groups, Actions, and Orbit Spaces
22. LCH Group Quotients, Hausdorffness, Connectedness, and Continuous Maps to Discrete Spaces

23. Connected Components, Closures, Path Connectedness, and Connectedness
24. Path Components, Local Path Connectedness, Partitions of Unity, Paracompactness, and Embeddings
25. Sigma-Compactness, Local Compactness, Paracompactness, and Normality
26. Partitions of Unity, Paracompact Manifolds, and Second Countability
27. Homotopy, Homotopy Classes, Composition, and Categories
28. Isomorphisms, Homotopy Equivalence, Functors, and Groupoids
29. Fundamental Groupoids, Fundamental Groups, Convex Spaces, and Pair Groupoids
30. Fundamental Groupoid Functor, Natural Transformations, and Homotopy
31. Natural Isomorphisms, Category Equivalence, and Full, Faithful, Essentially Surjective Functors
32. Category Equivalences, Fully Faithful Functors, Essential Surjectivity, and Pushouts
33. Pushout Uniqueness, Covers, Fundamental Groupoids, and Brown-Seifert-van Kampen
34. Fundamental Groupoid and Fundamental Group of the Circle
35. Proof of Brown-Seifert-van Kampen and Free Products of Group
36. Amalgamated Free Products and the Seifert-van Kampen Theorem
37. Degree of Circle Maps, Fundamental Theorem of Algebra, Compact-Open Topology
38. Compact-Open Topology
39. Uniform Convergence on Compact Sets and the Compact-Open Topology
40. Stone-Čech Compactification

11 Complex Analysis

1. Introduction, Complex Numbers, Holomorphic Functions, and Cauchy-Riemann Equations [text](#)
2. Power Series and Hadamard's Formula
3. Integration on Curves and Goursat's Theorem
4. Cauchy Integrals, Liouville's Theorem, and Fundamental Theorem of Algebra
5. Analytic Continuation, Moreira's Theorem, and Residue Calculus [text](#) [text](#)
6. Riemann Sphere, Argument Principle, Open mapping theorem, and Maximum Modulus [text](#) [text](#)
7. Homotopy, Complex Logarithm, Fourier Series, and Mean Value Property [text](#) [text](#)
8. The Fourier Transform and Poisson Summation [text](#) [text](#)
9. Trace Formula, Paley-Weiner, and Phragmen-Lindelof [text](#) [text](#)
10. Conformal maps and Schwarz lemma [text](#) [text](#)
11. Schwarz lemma, Automorphisms of the Disk and Half-Plane, and Riemann Mapping Theorem [text](#) [text](#)
12. Montel's Theorem (Arzela-Ascoli), Explicit Riemann Mapping Theorem, Dirichlet Problem, and Grotzsch's Thm, quasiconformality [text](#) [text](#)
13. Basel Problem, Riemann Zeta Function, Euler Product, Sum Reciprocals of Primes Diverges, Jensen's Formula, and Hadamard Factorization [text](#) [text](#)
14. Zeta Function, Gamma Function, Mellin Transform/Inverse, Perron's Formula, Skewes's Number, Analytic Continuation, Functional Equation for Zeta, Riemann's Explicit Formula, and Legendre-Gauss Conjecture [text](#) [text](#)
15. Nonvanishing of Zeta on 1-Line and Prime Number Theorem [text](#)
16. Elliptic Integrals, Functions, Double Periodicity [text](#)
17. Weierstrauss P-function, Elliptic Curve and Eisenstein series, modularity, BSD [text](#) [text](#)
18. Jacobi Theta Function, Sums of Two Squares [text](#)

12 Matrix Lie Groups

- | | |
|--|--|
| 1. Complex and Quaternions A | 12. What is the Logarithm B |
| 2. Complex and Quaternions B | 13. On the Derivative of a Homomorphism |
| 3. More Quarternions, Group Theory A | 14. Adjoint and Lie Algebra A |
| 4. More Quarternions, Group Theory B | 15. Adjoint and Lie Algebra B |
| 5. Chapter 2 and Isometries over $\mathbb{R}, \mathbb{C}$ and $\mathbb{H}$ A | 16. Lie Algebra Example 2nd Order BCH A |
| 6. Chapter 2 and Isometries over $\mathbb{R}, \mathbb{C}$ and $\mathbb{H}$ B | 17. Lie Algebra Example 2nd Order BCH B |
| 7. Path Connectedness for $SO(n)$ and $SU(2)$ A | 18. BCH identity A |
| 8. Path Connectedness for $SO(n)$ and $SU(2)$ B | 19. BCH identity B |
| 9. Lie Groups and Tangent Space at I A | 20. BCH calculation A |
| 10. Lie Groups and Tangent Space at I B | 21. BCH calculation B |
| 11. What is the Logarithm A | 22. Universal Covers and Lie's Theorem A |
| | 23. Universal Covers and Lie's Theorem B |
| | 24. Group Orbits and Homogeneous Space A |
| | 25. Group Orbits and Homogeneous Space B |

13 Functional Analysis

1. Linear Spaces: Definition, Examples and Linear Span
2. Linear Spaces: Quotient Spaces and Convex Sets
3. Normed Linear Spaces: Definition and Basic Properties
4. Completing a Normed Linear Space
5. Finite Dimensional Linear Spaces
6. Examples of Normed Linear Spaces
7. In Infinite Dimensions the Unit Ball is not Compact
8. Zorn's Lemma
9. The Hahn-Banach Theorem
10. Convex Sets and Gauge Functions
11. Geometric Hahn-Banach Theorems
12. Dual of a Normed Linear Space
13. Extension of Bounded Linear Functionals, Closed Linear Spans
14. Reflexive Spaces
 - (a) The Dual Space of $C([a,b])$
 - (b) An Application of the Hahn-Banach Theorem: the Moment Problem and Chebyshev Approximation
 - (c) A Dual Variational Problem in Optimal Control
 - (d) An Application of the Hahn-Banach Theorem: the Existence of a Green Function
15. Hilbert Spaces
16. Closed Convex Subsets of a Hilbert Space
17. Riesz and Lax-Milgram Representation Theorems
18. Orthonormal Sets and Closed Linear Spans
19. Orthonormal Bases
 - (a) A Quadratic Variational Problem

- (b) The Dirichlet Principle
- (c) Generalized Derivatives and Sobolev Spaces
- 20. Uniform Boundedness Principle
- 21. Weak Convergence
- 22. Uniform Boundedness of Weak Converging Sequences
- 23. Weak Sequentially Compactness
- 24. Weak* Topology
- 25. Applications of Weak Convergence
- 26. Bounded Linear Operators
- 27. Transpose of Bounded Linear Operators
- 28. Strong and Weak Convergence of Operators
- 29. Principle of Uniform Boundedness for Maps and Compositions
- 30. Open Map Principle
- 31. The Closed Graph Theorem
- 32. Examples of Bounded Linear Maps: Integral Operators
- 33. Symmetric Operators
- 34. Eigenvalues of Compact Symmetric Operators
- 35. The Fredholm Alternative
- 36. An Application to Integral Operators
- 37. Materials

14 Algebraic Geometry

15 Differentiable Manifolds

16 Advanced Real Analysis

1. Normed Vector Spaces [A](#) [B](#)
2. Banach Spaces [A](#) [B](#)
3. Hilbert Spaces [A](#) [B](#)
4. Adjoint Operator [A](#) [B](#)
5. L_p Spaces on the Real Line [A](#) [B](#)
6. More On L_p And L_2 Spaces [A](#) [B](#)
7. More On Operators On L_2 [A](#) [B](#)
8. Approximation Theory [A](#) [B](#)
9. The Fourier Transform: [A](#) [B](#)
10. The Fourier Transform & Wavelets [A](#) [B](#)
11. Wavelets & Multiresolution Analysis [A](#) [B](#)
12. Wavelets & B-Splines [A](#) [B](#)
13. Special Functions & Diff. Equation [A](#)

17 Metric Differential Geometry

18 Partial Differential Equations

Undergraduate Level

1. [Introduction to PDEs](#)
2. [Method of characteristics and the transport equation](#)
3. [Deriving the wave and diffusion equations](#)
4. [Solving the wave equation](#)
5. [The d’Almbert formula and properties of the wave equation](#)
6. [The maximum principle for the diffusion equation](#)
7. [Properties of the diffusion equation](#)
8. [Solving the diffusion equation on the real line](#)
9. [Calculating solutions to the diffusion equation](#)
10. [Solving the non-homogeneous diffusion and wave equations](#)
11. [More on nonhomogeneous wave equation; Boundary Value Problems](#)
12. [Boundary value problems and sine Fourier series](#)
13. [Calculating Fourier coefficients](#)
14. [Background for theory of Fourier series](#)
15. [Complex Fourier series and different notions of convergence](#)
16. [Mean-square convergence of Fourier series Part 1](#)
17. [Mean-square convergence of Fourier series part 2](#)
18. [Pointwise convergence of Fourier series \(part 1\)](#)
19. [Pointwise convergence of Fourier series part 2; Intro to Laplace Equation](#)
20. [Properties of Laplace’s equation](#)
21. [Boundary value problem for Laplace’s equation](#)
22. [Dirichlet boundary value problem for Laplace’s equation on a disc](#)
23. [Consequences of Poisson’s formula](#)

- 24. [Introduction to the Fourier transform with applications](#)
- 25. [Application of Fourier transform to Schrodinger equation](#)

Graduate Level

1. Course Introduction: Transport Equations, Notation, Solutions, and Key Questions
2. Linear Transport Equations and Well-Posedness: Uniqueness and Continuous Dependence
3. Classification of Second-Order PDEs and Balance Laws with Transport Examples
4. Implicit Function Theorem and Characteristics Method for Transport Equations
5. General Method of Characteristics with Examples for First-Order PDEs
6. Characteristics Method, Shocks, and Wave Equation from Physical Derivation
7. Method of Characteristics: Example 3 and Further Applications
8. Wave Equation from String Vibrations and d'Alembert's Formula
9. Wave Equation: Continuous Dependence, Support, and Domain of Dependence
10. Wave Equation: Energy Conservation, Uniqueness, and Duhamel's Principle
11. Duhamel's Principle and Wave Equations in Higher Dimensions
12. Wave Equations in Higher Dimensions, Euler-Poisson-Darboux, and Heat Equation Introduction
13. Wave Equation in R^3 : Radial Solutions and Jump Discontinuities
14. Self-Similar ODEs and the Fundamental Solution of the Heat Equation
15. Heat Equation Solution Properties and Delta Function Behavior with Wave Comparisons
16. Heat Equation Solutions: Recap and Uniqueness in Bounded Domains and R^n
17. Heat Equation Problems, Duhamel's Principle, and Decaying Diffusion Equations
18. Fourier Transform: Definition, Examples, and Setup for the Maximum Principle
19. Maximum Principle: Proof and Remarks with Heat Equation Uniqueness and Stability
20. Inhomogeneous Heat Equation, Uniqueness Conditions, and Separation of Variables Introduction
21. Wave Equation in the Quarter Plane: Domain of Dependence, Solutions, and Energy
22. Heat Equation: Fourier Series Solutions, PDE Verification, and Initial Data Convergence
23. Convergence Tests, Heat Equation Solutions, Robin Boundaries, and Sturm-Liouville Introduction

24. Heat Equation with Robin Boundary Conditions and Eigenvalue Equation Analysis
25. Sturm-Liouville Theory: Self-Adjointness, Eigenvalues, and L^2 Convergence
26. Symmetric Sturm-Liouville Problems, Fourier Series Completeness, and Gibbs Phenomenon
27. Wave Equation on Inhomogeneous Strings and Uniqueness of Solutions
28. Forced Heat Equation: Setup and Solution via Separation of Variables
29. Laplace Equation: Examples and Symmetries
30. Fundamental Solutions of Laplace and Poisson Equations
31. Green's Identities, Mean Value Property, and Maximum Principle for Harmonic Functions
32. Laplace Equation: Maximum Principle, Uniqueness, and Separation of Variables
33. Poisson Kernel Derivation, Green's Functions, and the One-Dimensional Poisson Equation
34. Green's Functions: Properties, Boundary Behavior, and Test Functions and Distributions
35. Weak Maximum Principle for the Heat Equation: Theorem and Proof
36. Distributions, Shock Waves and Rankine-Hugoniot Conditions, and Green's Functions
37. Green's Functions for Poisson Equation and the Method of Images
38. Green's Functions on the Real Line and Unit Ball with Method of Images
39. Neumann Boundary Green's Functions and Nonlinear Waves: Burgers Equations and Shocks
40. Green's Function Problems and Applications
41. KdV Equation: Dispersion Relation, Solitons, and Phase Plane Analysis
42. Fisher and Bistable Equations: Traveling Waves and Nonlinear Dynamics
43. Sine-Gordon Equation: Practice Problems and Applications

19 Ordinary Differential Equations

1. Terminology, separation of variables, integral curves, OTs
2. Integrating factor technique, exact DEqns and closed condition
3. Exact equations, geometry and simple connectedness
4. Overview of Chapter 2, end of 1st order
5. Operator Calculus, polynomials in D
6. Differential operators, complex roots, complex exponential
7. Theory of n -th order linear ODEs
8. Undetermined coefficients, method of annihilators
9. 2nd LI solution formula, Retarded Springs and SHM
10. Resonant frequency derivation, variation of parameters
11. Cauchy Euler problem
12. Review for Boss Fight 1
13. Systems and Reduction of Order, Matrices begin
14. Row Reduction, Null Space, Eigensolutions introduced
15. Eigenvectors and decoupling, theory of system of ODEs, examples
16. Introducing matrix exponential, complex example, real example
17. Eigenvector solutions, generalized e-vector solutions begin
18. Matrix exponential, generalized e-vectors, k-chains, Cayley Hamilton
19. Matrix exponential solutions, variation of parameters
20. Coupled springs problems, energy analysis begins
21. Phase plane plots, critical points, energy analysis
22. Laplace Transforms
23. Inverse Laplace Transforms
24. Laplace transforms for discontinuous functions

25. Laplace transform technique examples
26. Dirac Delta and Laplace Transforms
27. Power Series Technique, overview of series method
28. Review for Boss Fight 2
29. Power Series Technique, examples
30. Singularities and Frobenius Method
31. Fourier Series
32. Heat Equation
33. Wave Equations
34. Laplace's Equation
35. Euler-Lagrange equations, one-dimensional problems, geodesics
36. Lagrangian Mechanics, speed fmlas, double pendulum, spherical bowl
37. Central force motion, wild methods, DEqns in hyperbolic numbers
38. Discussion of Final Boss Fight

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